P L D 2020 Lahore 24

Before Abdul Aziz Sheikh, J

ALI IMRAN---Petitioner

Versus

FOREST WILDLIFE AND FISHERY DEPARTMENT through Secretary and 3 others---Respondents

Writ Petition No.35700 of 2016, heard on 31st October, 2019.

(a) Punjab Wildlife (Protection, Preservation, Conservation and Management) Act (II of 1974)---

----Ss. 9 & 2(n)---Constitution of Pakistan, Arts.9 & 14---Public interest litigation---Protected animal---Restriction on hunting of protected animal---Protection of wild life---Endangered species---Black Bucks Deers---Scope---Petitioner sought direction to Provincial Government to implement steps for protection of the endangered species of "Black Bucks Deers", which per the contention of petitioner, were almost extinct---Validity---Black Bucks Deers were endangered species and protected from hunting under S. 9(ii) of the Punjab Wildlife (Protection, Preservation, Conservation and Management) Act, 1974 and were also mentioned in the Third Schedule of the said Act---Provincial Government, was therefore, required to protect and preserve Black Bucks Deers in compliance with the provisions of the Punjab Wildlife (Protection, Preservation, Conservation and Management) Act, 1974---Protection of Black Bucks Deers was not only the fundamental duty of the Government but also of every citizen to ensure implementation of Fundamental Right to Life guaranteed under the Constitution---High Court directed Provincial Government to implement recommendations made by the court-appointed Commission's report titled, "Black Bucks Commission Report dated 06.09.2019" and to ensure that it meets its obligations under all relevant laws for protection of Black Bucks Deers---Constitutional petition was disposed of, accordingly.

Province of Punjab v. Lal Khan 2016 SCMR 48 rel.

(b) Constitution of Pakistan--

----Arts.9 & 14---Security of person---Right to life and liberty of person---Right to life---Inviolability of dignity of man---Nature of the Fundamental Right to life---Scope---Right to life guaranteed under Arts.9 & 14 of the Constitution included right to live in a world that had an abundance of all species, not only duration of current generation but also for its progeny.

Province of Punjab v. Lal Khan 2016 SCMR 48 rel.

Sheraz Zaka for Petitioner.

Dr. Parvez Hassan, Senior Advocate Supreme Court, Head of Black Bucks Commission along with Asad Ahmad Ghani for Respondents.

Saqib Haroon Chishti, A.A.G. along with Lt. (R) Sohail Ashraf, D.G. wildlife for Respondents.

Date of hearing: 31st October, 2019.

JUDGMENT

ABID AZIZ SHEIKH, J.---This constitutional petition has been filed by way of public interest litigation for the protection of endangered species of black bucks and deer which is virtually getting extinct.

2. The main contention of the petitioner is that black bucks are protected animals under the Punjab Wild Life (Protection, Preservation, Conservation and Management Control) Act, 1974 (Act) but appropriate measures are not being taken to protect black bucks and deer, who are not only dying but also being hunted. The claim of the petitioner is that respondents are required to maintain proper measures for preservation and protection of black bucks and deer.

3. This Court vide order dated 26.06.2018 appointed a commission to be headed by Dr. Parvez Hassan, Senior Advocate Supreme Court (Black Bucks Commission) to submit report regarding condition of Black Bucks in Lal Suhanra, National Park Bahawalpur and the possibility of progressively releasing of Black Bucks in Cholistan Desert and to finally suggest measures which can be taken by the Government in order to ensure that no poaching of Black Bucks and other deer take place in the Cholistan Desert.

4. The Black Bucks Commission after various meetings, deliberations and spot visits, recommended a comprehensive plan for the reintroduction of Black Bucks in the Cholistan in its final report dated 06.09.2019. The relevant measures and suggestions for reintroduction of the Black Bucks as recommended by the Black Bucks Commission in its final report are as under:-

> 2. Reintroduction of Blackbuck
>
> (1) The reintroduction of blackbuck is possible and is recommended. But it entails many challenges and reintroduction should only be undertaken if the challenges are first and duly met.
>
> (2) The reintroduction should be planned as a `conservation reintroduction' which also entails restoring the ecological function of the species. This planning should also be engaged or at least have a review by international experts either from the IUCN Reintroduction Specialist Group and/or IUCN Antelope Specialist Group. It is recommended that the Punjab Forest, Wildlife and Fisheries Department organize a workshop to bring together relevant, small group of experts to devise a plan specific to blackbuck reintroduction.
>
> (3) The first and foremost aspect of any reintroduction planning is ensuring that the

original threats that had caused the decline of the species have been removed. In case of Pakistan, the two major factors were hunting and habitat degradation. The authorities have not been able to control the hunting of chinkara and nilgai in the wild and this is a reflection of the fact that factors that had caused the extinction of blackbuck still prevail.

(4) As the reintroduction of the blackbuck is an important recommendation, the Commission cautions that it has to be carefully planned, keeping in mind that implementation of a reintroduction project is largely a bottom up approach and, therefore, the involvement and ownership of local communities is the most important aspect. This is particularly relevant to those that have customary rights, e.g., livestock holders, indigenous people, and those dependent on the habitat in any manner such as wood collectors.

(5) A site-specific Conservation Action Plan for Blackbuck is needed and agreed between stakeholders including the Government, conservation organisations (such as WWF, IUCN, Deer Foundation International) and local community. A similar example is available in case of the Shuklaphanta Wildlife Reserve (2016 - 2020) in Nepal which is an exclusive plan for a particular site for the re-establishment of a wild population of the species.

(6) An important aspect of reintroduction is to understand the changes that might have occurred over time in the habitat. It is highly recommended to conduct a GIS based study to understand such aspects.

(7) The population of blackbuck in the National Park is highly inbred and all the blackbucks in the captive facilities have been sourced from the National Park. It is important that blackbucks, particularly female, are brought from other facilities on priority. The genetic fitness of the blackbuck used for reintroduction is crucial to the success of the reintroduction.

(8) The Punjab Wildlife and Parks Department should keep updating the record of all blackbuck in Punjab and other provinces along with the sex ratiuo. Data of blackbuck in individual ownership is not available as the registered private breeding farms also sell animals to individuals and monitoring system by the Punjab Wildlife and Parks Department needs improvement. The trophies from expired animals can potentially end up in illegal wildlife trade.

(9) The Punjab Wildlife and Parks Department can organize private breeding farm owners to support blackbuck conservation breeding and maintain database to sustain a genetically diverse population through exchange of animals. There can be some incentives for this venture.

(10) RD-65 Deer Enclosure in the National Park originally housed all the population

of blackbuck in the National Park. But over time, the fence of this Enclosure, being old and depleted at number of places, was not considered secure to house the blackbucks due to which the animals were shifted to a smaller enclosure of 1 Km x 1 Km (1000 m x 1000 m) constructed inside the same RD 65. This space is insufficient for the animals being reared for release to the natural habitat. It is recommended that all female blackbucks in the small enclosure of RD 65 be shifted to the Houbara and Deer Foundation Enclosure (15 Kms x 4 Kms) in the National Park and the surplus male blackbucks from this Enclosure may be provided to the zoos, wildlife safari parks, and government-owned or private wildlife farms. The facility managed by the Houbara and Deer Foundation, at present, has a skewed sex ratio - 96 males and 38 females. The breeding potential survival and fitness of the population will enhance if females outnumber males. The Punjab Wildlife and Parks Department can support this by the proposed transfer of females from RD 65 to the Houbara and Deer Foundation Enclosure. Females could also be acquired from others facilities including through imports as indicated in Section H. 2(b) above.

(11) The blackbuck lives in groups, never solitary, mix sex groups 5- 20 animals and even bachelor groups. Single sex groups have been successfully formed for the blackbuck in captivity, removing young from the group earlier on similar arrangements can be made to manage correct mixed sex groups in the National Park.

(12) Comprehensive veterinary care and vaccination programme for the livestock will be needed around the habitat in the National Park finalised for reintroduction of blackbuck and should begin immediately to ensure that there is no loss of a reintroduced population due to disease transmission from livestock

3. Anti-Poaching and Predator Control

(1) An effective implementation of the 1974 Punjab Wildlife Act and the proposed Punjab Protected Areas Act, 2019, with the better-resourced and well-trained staff of game wardens, reinforced by "deterrent" penalties and backed by the support of local communities, all recommended by the Commission, will provide the best safeguards against poaching of the reintroduced blackbuck in the Cholistan Desert.

(2) The habitat areas of the blackbuck need to be effectively monitored for anti-poaching activities with the co-operation between the Army, law enforcing agencies and civil society.

(3) Culling of stray dogs will be required as they hunt on the calves/fawn of ungulates in the wild. A linked aspect is availability of food for the stray dogs which will require a proper management of solid waste.

(4) Jackals continue to be the dominant predator. A population survey of jackals is essential for any recommendation in respect of their control.

4. Other Suggestions

(1) The Government should ensure that the protected areas under the proposed Punjab Protected Areas Act, 2019, are not reduced or shrunk, directly or indirectly. In fact, there is a need to increase and expand it supporting the Convention on Biological Diversity (CBD).

(2) Deformed blackbucks should be euthanized using the most humane practice which usually includes darting the animal with an anesthetic agent and then giving IV dose of a euthanasia ' agent (American Veterinary Medical Association, 2013). It is not recommended, and it is considered unethical to have trophy hunting for those animals which are heavily domesticated and which do not run to save themselves. Further, hunting of any animal inside a fenced area is not defined as trophy hunting by the IUCN and it is referred as 'canned hunting' (IUCN, 2016). Trophy hunting should, accordingly, be conducted as a community based programme as per the IUCN Guidelines (IUCN/SSC., 2012), Annexure 12. It should not be allowed unless a viable/sustainable population has been established in the wild. Trophy hunting may, however, be allowed for surplus male blackbuck from the Houbara and Deer Foundation Enclosure.

(3) Any trophy hunting permit should be subject to a thorough population survey and monitoring and adaptive management. If any adverse impacts are seen, it must be stopped and if population continues to grow, trophy hunting of carefully selected animals can continue.

(4) There should be a notified trophy hunting license fee (for international and local hunters) as is the case with other ungulate species in Pakistan and it should be reviewed annually. The trophy hunting should be led as a programme of the Government with NGOs and local community. This would also require registration or nomination of the suitable community organizations. In case of international hunters, the CITES Management Authority of Pakistan must be fully involved.

(5) The experience under the Punjab Urial Conservation, Protection and Trophy Hunting (Committees) Rules, 2016, should be useful in the economic incentives provided to the local communities. The Commission also suggests that trophy hunting of the blackbuck should not be the only incentive for the communities; there should be more opportunities for them in employment related to conservation interventions and/or tourism related economic support.

(6) The basic function of a captive animal facility is conservation awareness of masses. The educational potential of the National Park has not been fully utilised. There should be information signs of international standards, education officers, restoration of the Patisar lake, and with it, a bird watching hut.

(7) There should be a short, professionally made (10-15 minutes) documentary on the National Park and Cholistan Game Reserve.

(8) Appropriate amendments should be made in the 1974 Punjab Wildlife Act in support of the recommendations of the Commission. It is essential to involve the local communities in Cholistan in the conservation and reintroduction of the blackbuck. The amendments should also accommodate community conservation areas which would support future post-release conservation efforts of blackbuck. There is also a need to increase penalties for illegal hunting.

(9) Establish a Blackbuck Research Unit in the Cholistan University of Veterinary Sciences or the Islamia University, Bahawalpur to enhance the research efforts on blackbuck in Cholistan.

(10) The National Park should be restored and managed in the true spirit of a UNESCO biosphere reserve.

(11) Globally, many protected areas are now managed using SMART - Spatial Monitoring and Reporting Tool. This tool helps in measuring, evaluating and improving the effectiveness of wildlife laws, patrols and site-based conservation activities. It is open sourced software and some trainings have already been done in Pakistan. Applying this in the National Park can effectively improve the management and protection of the area and this can be spread to the Cholistan Desert to protect other species such as chinkara.

J. Essential Requirements for Blackbuck Reintroduction

I. Planning and community stewardship

Planning and having all the stakeholders on board including local communities is pivotal to any conservation project. Community ownership needs to be established and local people should become stewards of blackbuck conservation. A share in trophy hunting should not be the only incentive for the local communities because trophy hunting should not be allowed unless a population has been established in the wild. An important indicator of a successful reintroduction project is breeding of the reintroduced population. Other incentives may include local employment, improvement of livestock health, community development initiatives, tourism, night safari and camping engagement communities etc.

2. Mitigating threats and zonation of the National Park and Cholistan Game Reserve

The IUCN Guidelines emphasize that threats that led to the extinction should be eradicated, illegal hunting and poaching of fawn was a serious issue and this needs to be managed and chinkara hunting can be indicative of effective law enforcement.

Joint management of different Departments is needed and a Management Plan for the Lal Suhanra National Park and Cholistan Game Reserve needs to be formally developed and budgeted for and reintroduction can begin with the National Park. The National Park needs a core area purely for biodiversity conservation; this zonation is also a requirement of a UNESCO biosphere reserve.

3. Management stratezies

Appropriate habitat selection for release is critical. There may be some possible management strategies needed which are based on monitoring of the released herd (which should be ideally satellite tagged). These can be variable, for example it may include supplementary feeding or provision of water or include fencing of agricultural areas to minimize conflict of introduced blackbuck with local communities. Chain-link fencing of a sizable habitat patch, where the animals seek daytime shelter, combined with other local protective methods in the cultivated areas would be helpful (Chauhan and Singh, 1990). Extreme measure seven include recapturing of released animals if the need arises, the safety of animals is foremost.

4. Selection of herd of release

The most important factor is having a genetically fit, compatible herd selected for release, and this must be acclimatized to the conditions of wild. In this respect the enclosure established by the Deer Foundation International is ideal. More understanding of the group structures within this enclosure is needed to determine the fitness for release.

The herd size for release should not be too small as it will have low breeding potential and more likelihood of becoming extinct. It should comprise of at least 30- 50 animals with a minimum 1:4 ratio of male to female.

5. Risk assessment:

The plan must incorporate all the likely risks to the reintroduction and strategize the remedial measures.

Likely risks to blackbuck reintroduction are tabulated below with suggested remedial measures. Usually such assessment is done in a workshop included in the Recommendations above.

Risk assessment

#	Risks	Mitigation
1.	Illegal hunting/ poaching of fawns	Patrolling teams, joint Management and inter departmental coordination. Capacity building of communities and establishing community conservation areas which are also encouraged under the Convention of Biological Diversity.

2	Predator prey Interaction	Unnatural predators e.g stray dogs near human inhabitation, should regularly culled and dumping sites of waste managed. Management decision can be taken for common wild predators e.g jackals
3	Disease outbreak/ transmission from livestock	The animals to be Reintroduced should be vaccinated. Livestock veterinary care improvement and restriction in free grazing
4	Competition with other ungulates in the habitat sharing the same ecological niche e.g chinkara	Selection of habitat which does not overlap with other ungulate, monitoring of reintroduced animals through satellite tags so study the range and if they are moving to areas with either more livestock/poor habitat quality etc.
5	Inadequate resources and capacity	This can be serious and a constant monitoring of funds and opportunities for more fundraising would be needed. It will be good to involve corporates.
6	Inadequate transparency	Everything needs to be transparent and information should be available. This is also listed in the IUCN Reintroduction Guidelines.
7	Drought	Supplementary feeding and artificial waterholes
8	Conflict with communities	Fencing of agriculture areas or possibly compensation. Community stewardship and incentives.

6. Funding and Resources

The Punjab Wildlife and Parks Department has already set up a Blackbuck Conservation Fund' which currently comprises of the trophy hunting license fees generated by the Deer Foundation International.

The Punjab Wildlife and Parks Departments has also developed a project for reintroduction. It must be borne in mind that a reintroduction programme is resource intensive and requires long term commitment. Fundraising for the blackbuck through internationally donor organizations will be challenging as these organisations are more geared towards supporting projects that involve conservation of those species that are already threatened in the wild. Considering that the blackbuck is globally in the category of 'Least concern', it is unlikely to receive attention of donor organizations such as GEF, Disney Conservation Fund, and The Whitley Fund. This would require more self sustaining ventures which can include ecotourism, integration of some conservation fee into the Cholistan cultural and sports events, funds generated through ticket money of visitors to captive facilities in the National Park.

K. IUCN Reintroduction Guidlines

The IUCN Reintroduction Guidelines provide the most comprehensive and well researched document for any reintroduction planning. The account below summarises these Guidelines to comprehend that reintroduction requires thorough planning and it is not a measure that should be taken lightly.

Conservation translocation is the premeditated movement of organisms which provides conservation benefits to the whole ecosystem. There are different types of translocations, one of which is reintroduction. Reintroduction involves the release of species into areas where they previously ranged, in the absence of conspecifics. Its aim is to establish a sustainable population of the principal species within their native region.

Reintroduction of species, though aids the process of conservation, needs justification. A feasibility assessment should be conducted which must include a balance of expected benefits against the risks. Moreover, it must be done only when there is substantial evidence about the absence of threats that led to the previous extinction. Hence, presence of high degree of uncertainty inhibits reintroductions and other types of conservation translocations. Even though reintroduction may seem a favorable option, former natural habitats may no longer be suitable. Therefore, returning the species to the place where they were found, may not be suitable.

Once the project of reintroduction is conceptualized, its design and execution should follow the standard stages of a project design and management which includes gathering baseline information, analysis of threats and regular monitoring. A project based on these stages will help to ensure that the habitat meets all the biotic and abiotic requirements of the species. Whereas the process of monitoring will allow identification of new threats, leading to subsequent adjustments. Most importantly, human interests must not be ignored throughout the project and social, economic and political factors must be kept under consideration.

The whole process of translocation needs to be recorded and the public should have access to the results. This would help in future conservation projects and create awareness within the society.

* Translocation is an effective conservation tool but its use either on its own or in conjunction with other conservation solutions needs rigorous justification. Feasibility assessment should include a balance of the conservation benefits against the costs and risks of both the translocation and alternative conservation actions.

*	Attempts to reintroduce a species, if poorly conceived or implemented, may actually obscure the conservation issues that led to the decline of the species in the first place - and thus may detract from, rather than add to, a species chances of survival. (IUCN 1987)

*	There should generally be strong evidence that the threat(s) that caused any previous extinction have been correctly identified and removed or sufficiently reduced.

*	Any conservation translocation should follow a logical process from initial concept to design, feasibility and risk assessment, decision-making, implementation, monitoring, adjustment and evaluation.

*.	Risk is the probability of a risk factor occurring, combined with the severity of its impact. Individual risks will generally increase as the following increase in scale:

	1. The duration of any extinction period,

	2. The extent of ecological change during any extinction period,

	3. The degree of critical dependence of the focal species on others,

	4. The number of species to be translocated,

	5. The genetic differences between the original form and the translocated individuals,

	6. The potential negative impacts on human interests,

	7. The probability of unacceptable ecological impacts,

	8. Whether the translocation is into or outside indigenous range.

*	Many aspects of the translocated organisms' biology are relevant to the release strategy.

*	The age/size, sex composition and social relationships of founders may be optimised for establishment and the population growth rate stated in the objectives,

*.	Translocation success increases with the numbers of individuals released (which is often enhanced through multiple release events across more than one year),

*	The monitoring programme is the means to measure the performance of released organisms against objectives, to assess impacts, and provide the basis for adjusting objectives or adapting management regimes or activating an exit strategy.

	Adequate resources for monitoring should be part of financial feasibility and commitment.

*.	The translocation of captive-bred animals should follow a precise four-step protocol: feasibility study, preparation phase, release phase, and monitoring phase.

*.	The declining or extinction factors should be highlighted and removed before release into the wild.

*. It is highly recommended that the release sites be prepared using habitat management techniques to maximize the survival probabilities.

*. The release must be preceded by accurate veterinary control.

*. A reintroduction plan can be considered successful only when the following points are satisfied: the offspring of the founders start to breed, a minimum viable population (MVP) is reached and maintained, and the recruitment rate is higher than the adult death rate for three years.

*. Captive breeding is often used together with translocation projects, such as reintroduction and population reinforcement, but it must be stressed that these kinds of measures need to be carefully projected, because they could have many unexpected negative effects and should be always subordinated to habitat conservation.

The projects should involve the preparation and education of local people.

*. The genetically and geographically closest populations should be chosen for breeding and translocation, to preserve genetic identity and homogeneity.

* It could be advisable to vaccinate the founders to limit the mortality due to local diseases.

*. In the wild, individuals with superior abilities stand the greatest chance of long-term survival, reaching sexual maturity and passing their genes on to future generations. In captivity, most animals,- receive total institutional care. They do not require the same skills and physical traits that their wild counterparts require. Undesirable attributes may be carried forward from generation to generation because natural selection is not playing a part (emphasis added).

L. Reintroductions of Ungulates Globally

There are numerous successful reintroductions of ungulates globally. One of the most iconic is perhaps of the Przewalski's horse. Mongolia's last group of takhi (Przewalski's horse) was spotted around 1969. It took another 20 years for conservation and breeding programs to become effective and for the horse to show signs that it might survive. By 1990, the population had reached nearly a thousand, with 961 P-horses living in over 129 institutions in 33 countries-enough to try reintroducing the takhi to the wild. All of today's reintroduced takhi descend from just 12 captured horses and several cross-breeding.

Some of the horses could not be released into the wild directly from zoos-the animals needed a "semi-reserve" area, a sort of base camp in the form of a fenced enclosure, for acclimatization.

After a 2-year period of captive management, the first herd of 10 animals was released to the wild on 31 January 1982.

5. The black buck deer is endangered species, protected under the Act. Section 2(n) of the Act provides the definition of protected animals and section 9 (ii) of Act put a restriction on the hunting of protected animals. In the third schedule of the Act at serial No.67 of 2nd category of mammals, the black buck (Antelope Cervicopra) is mentioned as protected animal. Therefore, the Provincial Government is required to protect and preserve black buck deer in compliance of provision of the Act.

6. The Hon'ble Supreme Court of Pakistan in case of Province of Punjab v. Lal Khan (2016 SCMR 48) observed that the right to life guaranteed under Articles 9 and 14 of the Constitution of Islamic Republic of Pakistan, 1973 (Constitution) includes right to live in a world that has an abundance of all species not only for the duration of our lives but available for our progeny too. It has now been scientifically established that if the earth becomes bereft of birds, animals, insects, trees, plants, clean rivers, unpolluted air and soil, it will be the precursor of our destruction/extinction. The black buck deer is one of the endangered species and protection of such species is not only fundamental duty of government but of every citizen to ensure right to life guaranteed under Constitution.

7. The learned Law Officer has placed on record the written instructions from the Government of Punjab (Forestry, Wild Life and Fisheries Department) and stated that the Department endorses the above Black Bucks Commission Report dated 06.09.2019 containing certain recommendations for the protection of endangered species of Black Bucks. The learned counsel for the petitioner also submits that this writ petition may be disposed of in terms of the Black Bucks Commission Report dated 06.09.2019.

8. In view of above discussion and also for consensus developed between the parties, this petition is disposed of in terms of the recommendations made by the Black Bucks Commission in its report dated 06.09.2019, for the protection of the endangered species of Black Bucks. The Provincial Government is further directed to ensure that the obligations under the relevant laws including the Act are fulfilled for protection of Black Buck. To ensure the implementation of the recommendations made in the Black Bucks Commission Report, it is also directed that Black Bucks Commission shall meet once every six months for next three years and report to this Court through the Deputy Registrar (Judicial) on the progress and implementation of Blacks Bucks Commission's Report.

9. Before parting with this judgment, this Court must place on record its appreciation for the hard work untaken by the Black Buck Commission headed by Dr. Parvez Hassan, Senior Advocate Supreme Court. This Court also appreciates the effort of the petitioner and his learned counsel for highlighting this important issue in the public interest litigation.

KMZ/A-91/L Order accordingly.